

QualiPharm Inc. — Site A (US Operations)
STANDARD OPERATING PROCEDURE
Packaging Integrity and Seal Testing

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Jurisdiction	US — Site A (FDA 21 CFR Part 211)
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Classification	ACTIVE — Current Version

1. Purpose

This SOP defines the methods, acceptance criteria, and failure actions for packaging integrity and container closure integrity testing (CCIT) of finished pharmaceutical products at QualiPharm Site A. Packaging integrity is critical to product quality, patient safety, and compliance with FDA 21 CFR Part 211.132 (tamper-evident packaging) and 21 CFR Part 211.94 (drug product containers and closures).

2. Scope

Applicable to:

- All solid oral dosage forms: tablets, capsules, caplets in bottles or blister packs
- All semi-solid products in tubes or jars
- Packaging integrity testing during batch release and in response to quality complaints
- Complaint investigation for packaging-related defects (complaint_type: quality or contamination)

3. Seal Test Methods

3.1 Vacuum Decay Method (Primary — Bottles)

Applicable to HDPE and glass bottles with induction-sealed or screw-cap closures.

1. Place sealed bottle in vacuum chamber
2. Apply vacuum to 10 kPa absolute pressure
3. Hold for 30 seconds
4. Measure pressure decay using calibrated transducer
5. Acceptance criterion: pressure decay $\leq$ 0.5 kPa over 30 seconds

3.2 Dye Ingress Method (Bottles and Blister)

Used for confirmatory testing when vacuum decay is inconclusive.

6. Immerse container in 0.1% methylene blue solution
7. Apply 10 kPa vacuum for 30 minutes
8. Remove container, rinse exterior
9. Open container and examine interior and product surface
10. Acceptance criterion: no dye detected on product or interior surfaces

3.3 Burst / Peel Test (Blister Packs)

Evaluates seal integrity between blister foil and backing card.

11. Sample 10 blister strips per batch at start, middle, and end of packaging run
12. Mount blister strip in tensile testing machine
13. Apply peel force at 300 mm/min
14. Record peak peel force and seal width
15. Acceptance criterion: peel force ≥ 8 N/15mm width, uniform peel pattern with no delamination

3.4 Visual Inspection (All Formats)

100% in-process visual inspection plus AQL sampling at batch release.

- Seals must be uniform, complete, and free of channels or leaks
- Induction seals must show complete foil-to-bottle bond with no lifted edges
- Blister foil must show no pin-holes, cracks, or incomplete seals visible by transmitted light
- Acceptance Level: AQL 0.65 for critical defects (seal breach), AQL 1.0 for major defects (incomplete seal)

4. Acceptance Criteria Summary

Test Method	Container Type	Parameter	Acceptance Criterion
Vacuum Decay	Bottles	Pressure Decay	≤ 0.5 kPa / 30 sec
Dye Ingress	Bottles / Blisters	Dye Penetration	No dye detected
Peel / Burst	Blister Packs	Peel Force	≥ 8 N/15mm
Visual 100%	All formats	Seal integrity	No defects per AQL 0.65

5. Failure Actions

5.1 Single Unit Failure

16. Isolate failed unit immediately
17. Extend sample size to 2x AQL quantity
18. If extended sample passes — release with documented deviation
19. If extended sample fails — proceed to batch failure actions

5.2 Batch Failure Actions

- 20. Quarantine entire batch — affix QUARANTINE label
- 21. Notify QA Director and Batch Record Review team within 1 hour
- 22. Initiate complaint investigation in QualiTrace (complaint_type = quality)
- 23. Review packaging line parameters: seal temperature, dwell time, pressure, foil lot
- 24. Perform root cause analysis — document in investigation report
- 25. Do not release batch until root cause identified and CAPA initiated

5.3 Complaint-Triggered Testing

When a packaging integrity complaint is received (seal failure, moisture ingress, contamination due to seal breach):

- 26. Retrieve retained samples from same batch_number
- 27. Perform full CCIT testing using all applicable methods above
- 28. Document results in QualiTrace investigation report as evidence_summary
- 29. Compare to original batch release data to identify if failure occurred during distribution or manufacture

6. Revision History

Version	Date	Author	Summary
2.1	Mar 2024	QA/Pkg Eng	Updated acceptance criteria. Added complaint-triggered testing section.
2.0	Jun 2022	QA Dept	Added vacuum decay method as primary test. AQL updated to 0.65 for critical.